Rivindu Vinsara Kumarage

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Professional Summary

Electronic & Telecommunication Engineering undergraduate at the University of Moratuwa building robotics software in C++ and Python: ROS 2 navigation, SLAM, perception and teleoperation across simulated and physical robots. Completed a 27-week software engineering internship shipping containerised services with Docker and Kubernetes. Comfortable debugging the full stack, from STM32 firmware up to ROS 2 autonomy.

Education

University of Moratuwa — **Moratuwa, Sri Lanka**
Bachelor of Science in Electronic & Telecommunication Engineering, ***Current GPA: 3.71/4.00*** — *2022–2027*
Relevant Coursework: Robotics, Autonomous Systems, Control Systems, Deep Learning, Image Processing & Computer Vision, Embedded Systems, Pattern Recognition, Data Structures & Algorithms, Signal Processing, Linear Algebra

ESOFT — **Sri Lanka**
Level 3 Diploma in IT & Diploma in English, ***Both completed with Distinction*** — *2022–2023*

Nalanda College — **Colombo 10, Sri Lanka**
G.C.E. Advanced Level, ***Z Score: 2.26*** — *2013–2022(Jan)*
Results: Combined Mathematics (A), Physics (A), Chemistry (A), General English (A)

Experience

Yaala Labs (Pvt) Ltd — **Colombo, Sri Lanka**
Software Engineering Intern, 27 weeks — *Nov 2025 – May 2026*

- Developed and shipped backend services in TypeScript/Node.js and Java for **P8**, a live production fintech platform, through Git-based code review and end-to-end test suites.
- **Containerised and deployed** services with **Docker, Podman and Kubernetes** on AWS (EC2, ECS Fargate, Lambda), and investigated service failures and production deployment issues on **Arch/Linux**.
- Led the transition from Auth0 to **self-hosted Keycloak**, implementing PKCE login, token validation and AWS Lambda authorizers.
- Added feature-flagged support for both providers, allowing deployments to switch between Auth0 and Keycloak during the transition.
- Connected blockchain and custody Vert.x services through a **Hazelcast cluster** and event bus for cross-pod request routing.

Technical Skills

Languages	:	C++, Python, C, Bash | **Familiar:** MATLAB, TypeScript, Verilog, SQL
Robotics	:	ROS 2 (Humble), rclcpp, rclpy, Gazebo, RViz, MuJoCo, Isaac Sim
Navigation & SLAM	:	A*, EKF-SLAM, KISS-ICP, RTAB-Map, occupancy grids, obstacle inflation, PID control
Perception	:	OpenCV, Open3D, RGB-D (ZED 2i), point clouds, ArUco/QR
Embedded & Hardware	:	STM32, Raspberry Pi, ESP32, IMU/encoders, LiDAR, UART/I2C/SPI, BLE
DevOps & Systems	:	Arch/Linux, Git, Docker, Podman, Kubernetes, AWS, CI/CD
ML	:	PyTorch, TensorFlow

Key Projects

Final-Year Project — Scene-Graph-Based Navigation & Dynamic Sensing (Unitree Go2)
Python, C++, ROS 2 Humble, PyTorch, Open3D, MuJoCo, Isaac Sim, Gazebo, ZED 2i

- Extended DovSG's navigation pipeline with robot-height occupancy-grid extraction, obstacle inflation by robot radius, A* planning and heading-aware waypoints; added 21 automated tests, bringing the suite to 84 passing.
- Developed a **ROS 2 waypoint-following controller** (rclpy) with odometry feedback, rotate-then-drive PID control and velocity commands for quadruped navigation.
- Implemented a native **C++ (rclcpp) sensor bridge** for a MuJoCo Go2, publishing 4,000-ray LiDAR at 10 Hz, IMU and simulation clock, with ROS 2 teleoperation and KISS-ICP mapping.
- Generated **3D point-cloud maps in MuJoCo, Isaac Sim and Gazebo** as testbeds for the same final-year project, using LiDAR/RGB-D sensing with KISS-ICP, DROID-SLAM and RTAB-Map.
- Transferred ZED 2i SVO recordings from a **Jetson Nano** to the ENTC server over **Tailscale** for mapping; fused frames into a **1 cm voxel map** with DROID-SLAM poses, floor alignment and coloured ICP refinement.

Repos: isaac-go2-ros2 · unitree_mujoco · go2_ros2_sim_py · zed_mapping_check

2D EKF-SLAM — Localisation & Mapping
Python, NumPy, ROS 2, Gazebo Classic, RViz

- Implemented EKF-SLAM from scratch: prediction, range–bearing correction, correlated landmark initialisation, wrapped angle innovations, Joseph-form covariance updates and loop-closure diagnostics.
- Benchmarked 120 runs (6 scenarios, 20 paired seeds): trajectory RMSE 0.238 ± 0.146 m versus 3.233 ± 1.172 m for wheel odometry; added a Gazebo/ROS 2 demo and 11 unit tests.

PixieBot — Autonomous Manipulation & Sorting Robot
C++, PlatformIO, 4-DOF Arm, Encoders, PID

- Built control software for a 4-DOF sorting robot with encoder-PID navigation and colour recognition; 4th place, SLRC 2025. Related sensor-based robot: EN2533 design project.

Micromouse — 16×16 Autonomous Maze Robot
Embedded C, STM32F411, Altium, PID, Sensor Fusion

- Built firmware for a 16×16 micromouse — flood-fill exploration, shortest-path verification, IR wall sensing, MPU9250 gyro fusion, quadrature encoders and PID motion control — plus its Altium PCB and CAD parts; Robofest 2025 finalist.

Lunar Lander RL — PPO Landing Controller
Python, PyTorch, Stable-Baselines3, Gymnasium, PPO, TensorBoard

- Built a PPO landing controller for Gymnasium's LunarLander-v3 with Stable-Baselines3, TensorBoard logging, seeded runs and a locked dependency environment for reproducible training on CPU.
- Evaluated the policy against a random-action baseline, and compared default shaped rewards with a custom tilt-shaping wrapper under equal training budgets, with learning curves and recorded landing episodes.

Boxsmith — Object Detection & mAP Lab
Python, PyTorch, Streamlit, Computer Vision

- Trained a 173k-parameter CNN from scratch on synthetic shapes and built an interactive lab for bounding-box regression, NMS, IoU, precision–recall and mAP, with reproducible seeded datasets and CI-tested releases.

Shortcut Sleuth — CNN vs ViT Robustness Study
Python, PyTorch, Vision Transformers, Reproducible Research

- Ran a controlled 12-run, three-seed CNN vs ViT study on background shortcut learning with paired foreground-preserving swaps, measuring background flip rates and worst-group accuracy on 3,000 held-out objects.

Noise to Vogue — Diffusion Model Lab
Python, PyTorch, DDPM, FastAPI, CI

- Trained two unconditional DDPMs (3-resolution U-Net, 200 timesteps, 8,000 updates each) on Fashion-MNIST under identical seeds and data order, isolating linear versus cosine noise schedules.
- Evaluated both on 1,024 held-out examples and shipped a FastAPI viewer for ancestral sampling, forward/reverse Markov chains and per-timestep noise prediction, with a pytest suite in CI.

Additional projects: AWS ECS (Terraform); School Management API (CRUD); PathoAssist; Echo-Guard; Zero-shot image–text retrieval; BLE torque sensor.

Awards & Competitions

SLRC 2025 — 4th place (PixieBot) — *university category finalist, Sri Lanka Robot Challenge.*

Brainstorm 2025 — 4th place (PathoAssist) — *finalist in an inter-university innovation competition.*

SPARK Challenge 2025 — 6th place (Echo-Guard) — *innovative, sustainable and socially conscious solution.*

Additional Information

Professional: **IEEE Student Member**; **E-Club Committee Member** — organised inter-university competitions

Languages: English (Fluent), Sinhala (Native)

Other: Taekwondo (UoM team, Green Belt 7th GUP); Arduino, MATLAB and Stanford ML (Coursera) certificates

References

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